AN OPEN SUBSCHEME OF A LOCALLY NOETHERIAN SCHEME IS SCHEME THEORETICALLY DENSE IF AND ONLY IF IT CONTAINS ALL EMBEDDED POINTS, IF AND ONLY IF IT CONTAINS ALL ASSOCIATED POINTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The canonical Stacks tag of the counted statement is 083P (divisors-lemma-scheme-theoretically-dense-contain-embedded-points). Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the tag is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The definition of embedded point is attached as a context span because the statement speaks of the embedded points of X without repeating the definition. The paragraph that precedes that definition in the source cites EGA and is therefore not included.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of sources/stacks-chow/preamble.tex lines 14-18 and 154-217. No macro was invented or redefined.

Allowed standard prerequisites (editorial).

- Scheme theoretically dense open subscheme. Morphisms, Definition labelled morphisms-definition-scheme-theoretically-dense (canonical Stacks tag 01RB); cited in the statement at line 446. Supplies the notion that condition (1) of the statement refers to; the proof of the equivalence is carried out inside the counted span.
- A quasi-compact open of a Noetherian affine scheme is a finite union of principal opens D(f_1) up to D(f_n). Algebra, Lemma labelled algebra-lemma-qc-open (canonical Stacks tag 00F6); invoked at proof line 457. Produces the elements f_1, ..., f_n whose localisations replace U throughout the proof.
- A map A -> A_{f_1} x ... x A_{f_n} is injective exactly when every associated prime of A misses at least one f_i. Morphisms, Example labelled morphisms-example-scheme-theoretic-closure (canonical Stacks tag 056C), used at proof line 460; and Algebra, Lemma labelled algebra-lemma-zero-at-ass-zero (canonical Stacks tag 0311), used at proof line 483. Translates scheme theoretic density into an injectivity statement and back; the two directions of the equivalence are then proved in the counted span.
- The associated primes of a Noetherian ring are the minimal primes together with the embedded associated primes. Algebra, Proposition labelled algebra-proposition-minimal-primes-associated-primes (canonical Stacks tag 02CE); invoked at proof lines 472-475. Identifies condition (2) with condition (3) once density is read as containing all minimal primes.

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Source reference. Original definition 01RB: sources/stacks-chow/tags.txt, source lines 2296–2296. Cross-chapter pointer for the notion appearing in condition (1) of the statement (cited at statement line 446); the equivalence itself is proved inside the counted spans.

Source reference. Original reference 056C: sources/stacks-chow/tags.txt, source lines 6705–6705. Cross-chapter pointer invoked at proof line 460 for the translation of scheme theoretic density into injectivity of A -> A_{f_1} x ... x A_{f_n}; both directions of that translation are then used explicitly in the counted proof.

Source reference. Original lemma 01OX: sources/stacks-chow/tags.txt, source lines 2210–2210. Cross-chapter pointer invoked at proof line 455 for quasi-compactness of an open immersion into a Noetherian scheme; finiteness of the cover D(f_1) union ... union D(f_n) is then produced and used inside the counted proof.

Source reference. Original lemma 00F6: sources/stacks-chow/tags.txt, source lines 560–560. Cross-chapter pointer invoked at proof line 457 for writing a quasi-compact open of Spec(A) as a finite union of principal opens; the resulting elements f_1, ..., f_n are used throughout the counted proof.

Source reference. Original proposition 02CE: sources/stacks-chow/tags.txt, source lines 3081–3081. Cross-chapter pointer invoked at proof lines 472-475 for the description of the associated primes as the minimal primes together with the embedded associated primes; the equivalence of conditions (2) and (3) is derived from it inside the counted span.

Source reference. Original lemma 0311: sources/stacks-chow/tags.txt, source lines 3968–3968. Cross-chapter pointer invoked at proof line 483 for injectivity of A into the product of its localisations at the associated primes; this is the last step of the counted proof and is used exactly as stated.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $U \subset X$ be an open subscheme. The following are equivalent*

(1) *U is scheme theoretically dense in X (Morphisms, Definition 01RB),*
(2) *U is dense in X and U contains all embedded points of X, and*
(3) *U contains all associated points of X.*

Proof. The question is local on X, hence we may assume that $X = \mathrm{Spec}(A)$ where A is a Noetherian ring. Then U is quasi-compact (Properties, Lemma 01OX) hence $U = D(f_1) \cup \ldots \cup D(f_n)$ (Algebra, Lemma 00F6). In this situation U is scheme theoretically dense in X if and only if $A \to A_{f_1} \times \ldots \times A_{f_n}$ is injective, see Morphisms, Example 056C. Condition (3) translated into algebra means that for every associated prime $\mathfrak{p}$ of A there exists an i with $f_i \notin \mathfrak{p}$.

Assume (1), i.e., $A \to A_{f_1} \times \ldots \times A_{f_n}$ is injective. Say $a \in A$ has annihilator a prime $\mathfrak{p}$. Since a maps to a nonzero element of A_{f_i} for some i we see that $f_i \notin \mathfrak{p}$. Thus (3) holds.

Note that U is dense in $\mathrm{Spec}(A)$ if and only if it contains all minimal prime ideals. Since the set of associated primes of A is equal to the union of the set of minimal primes and the set of embedded associated primes (see Algebra, Proposition 02CE), we see that (2) and (3) are equivalent.

Assume (3). This means that every associated prime $\mathfrak{p}$ of A corresponds to a prime of A_{f_i} for some i. Then $A \to A_{f_1} \times \ldots \times A_{f_n}$ is injective because $A \to \prod_{\mathfrak{p} \in \mathrm{Ass}(A)} A_{\mathfrak{p}}$ is injective by Algebra, Lemma 0311. In this way we see that (3) implies (1). □

Definition 0.2. Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent sheaf on X.

(1) An *embedded associated point* of $\mathcal{F}$ is an associated point which is not maximal among the associated points of $\mathcal{F}$, i.e., it is the specialization of another associated point of $\mathcal{F}$.
(2) A point x of X is called an *embedded point* if x is an embedded associated point of $\mathcal{O}_X$.
(3) An *embedded component* of X is an irreducible closed subset $Z = \overline{\{x\}}$ where x is an embedded point of X.